

The relationship between dietary protein intake and blood pressure: results from the PREMIER study.



Observational and clinical studies suggest that high protein intake, particularly protein from plant sources, might reduce blood pressure (BP). To examine the association of dietary protein with BP, we analysed data from PREMIER, an 18-month clinical trial (n=810) that examined the effects of two multi-component lifestyle modifications on BP. We examined the association of protein intake with BP, and in particular the independent relationship of plant and animal protein with BP. Multivariable linear regression analyses were performed with both cross-sectional and longitudinal data. Dietary plant protein was inversely associated with both systolic and diastolic BP in cross-sectional analyses at the 6-month follow-up ($P=0.0045$ and 0.0096 , respectively). Fruit and vegetable intake was also inversely associated with both systolic and diastolic BP cross-sectionally at 6 months ($P=0.0003$ and 0.0157 , respectively). In longitudinal analyses, a high intake of plant protein at 6 months was marginally associated with a reduction of both systolic and diastolic BP from baseline to 6 months only ($P=0.0797$ and 0.0866 , respectively), independent of change in body weight and waist circumference. Furthermore, increased intake of plant protein, and fruits and vegetables was significantly associated with a lower risk of hypertension at 6 but not at 18 months. Results of this study indicate that plant protein had a beneficial effect on BP and was associated with a lower risk of hypertension at 6 months. Our data, in conjunction with other research, suggest that an increased intake of plant protein may be useful as a means to prevent and treat hypertension.